



Mahindra University Hyderabad
École Centrale School of Engineering
End-Semester Examination

Program: B. Tech.

Branch: CM

Year: II

Semester: I

Subject: Signals and Systems (MA2105)

Date: 10/12/2025

Start Time: 10:00 AM

Time Duration: 3 Hours

Max. Marks: 100

Instructions:

- 1) Answer all the questions.
- 2) All questions are self-explanatory; no clarification will be provided during the exam.

1. (20 points) Choose the correct option for each question and justify your answer briefly.

(a) A Linear Time-Invariant (LTI) system is causal if:

(i) $h[n] = 0$ for all $n < 0$

(ii) $h[n] = 0$ for all $n > 0$

(iii) $\sum |h[n]| < \infty$

(iv) None of these

(b) The system defined by $y[n] = u[n + 3]$ is:

(i) Causal

(ii) Stable

(iii) Non-causal

(iv) None of these

(c) The Z-transform of $\delta[n - 3]$ is:

(i) z^{-3}

(ii) z^3

(iii) 1

(iv) 0

(d) The inverse Fourier transform of a constant 1 is:

(i) $\delta(t)$

(ii) $2\pi\delta(t)$

(iii) $u(t)$

(iv) None of these

(e) The Region of Convergence (ROC) of $x[n] = (-2)^n u[n]$ is:

(i) $|z| > 2$

(ii) $|z| < 2$

(iii) $|z| = 2$

(iv) None of these

(f) The Z-transform of the sequence $x[n] = \{1, 2, 3\}$ is:

(i) $1 + 2z^{-1} + 3z^{-2}$

(ii) $1 + 2z + 3z^2$

(iii) $3 + 2 + 1$

(iv) None of these

(g) The Fourier transform pair of $e^{-at}u(t)$ ($a > 0$) is:

(i) $\frac{1}{a+j\omega}$

(ii) $\frac{1}{a-j\omega}$

(iii) $\frac{1}{j\omega}$

(iv) $2\pi\delta(\omega - a)$

(h) For the differential equation

$$\frac{d^2 y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5x(t),$$

find the transfer function $H(s) = \frac{Y(s)}{X(s)}$.

(i) $\frac{5}{s^2 + 3s + 2}$

(ii) $\frac{2}{s^2 + 5s + 3}$

(iii) $\frac{s+3}{5s^2 + 2s}$

(iv) $\frac{5s}{s^2 + 2s + 3}$

(i) The formula for the N -point discrete Fourier transform (DFT) of a sequence $x[n]$ is:

(i) $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$

(ii) $X[k] = \sum_{n=0}^{N-1} x[n]e^{j\frac{2\pi}{N}kn}$

(iii) $X[k] = \int x(t)e^{-j\omega t}dt$

(iv) None of these

(j) To check whether a system T is linear, we verify:

(i) $T(ax_1 + bx_2) = aT(x_1) + bT(x_2)$

(ii) Time invariance

(iii) Causality

(iv) Stability

2. (20 points) Find the Z-transform and the region of convergence (ROC) of the following signals:

(i)

$$x[n] = \begin{cases} \left(\frac{1}{5}\right)^n, & n \geq 0, \\ (-4)^n, & n \leq -1, \end{cases}$$

(ii)

$$x[n] = u[n] - u[n-5].$$

3. (20 points) Solve the following problems:

(i) Find the Fourier Transform of $x(t) = e^{-at}u(t)$, where $a > 0$.

(ii) Determine $|X(\omega)|$, where $X(\omega)$ is the Fourier transform of $x(t)$.

(iii) Compute the signal energy in the time domain.

or

For $my'' + cy' + ky = f(t)$, using Laplace transform

(i) Find transfer function $H(s) = \frac{Y(s)}{F(s)}$.

(ii) Compute impulse response $h(t)$ for $m = 1, c = 3, k = 2$.

(iii) Find step response $y(t)$ when $F(t) = u(t)$.

where y' and y'' denote the first and second derivative of y with respect to t , respectively.

4. (20 points) Consider the signal

$$x(t) = \begin{cases} t, & 0 \leq t < 2, \\ 2, & 2 \leq t < 4, \\ 0, & \text{otherwise.} \end{cases}$$

- (i) Find the expression and sketch the signal $y_1(t) = 4x(t)$.
- (ii) Find the expression and sketch the signal $y_2(t) = x(t) - 2$.
- (iii) Find the expression and sketch the signal $y_3(t) = 2x(t) + 1$.

5. (20 points) Solve the following problems:

- (i) Verify whether the signals

$$x_1(t) = \sin(t), \quad x_2(t) = \cos(t)$$

are orthogonal over the domain $t \in [0, 2\pi]$.

- (ii) Find the sum of two signals $x_1(t) = u(t+1)$ and $x_2(t) = u(t-2)$.
- (iii) Find the Z-transform of $x[n] = \delta[n+1] + \delta[n-1] + \delta[n-2]$.
- (iv) Consider a discrete-time LTI system with impulse response

$$h[n] = \alpha^n u[n], \quad \alpha \in \mathbb{R}.$$

Is the system causal or noncausal?